



Teheran University of Medical Sciences

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Final Exam
International Campus , DDS,MD,PharmaD Program
Biochemistry 2, Metabolism, Basic biochemistry

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Course Coordinator: Dr. Panahi

Date of Exam: Jul. 13, 2024

Duration of Exam: 55 minutes

MCQ Question

52

Directions:

- Each of the following questions contains one correct answer only.
- Please choose the correct answer for the MCQs in the answer sheet with pencil.



- 2
G6M
1. Which of the following enzymes is required for the regeneration of reduced glutathione?
☐ a. Glucose 6 phosphatase
☒ b. Phosphoglucomutase
☐ c. Lactate dehydrogenase
☐ d. Glucose 6 phosphate dehydrogenase
 2. Pentose phosphate pathway is responsible to make all the following EXCEPT
☐ a. CO_2
☒ b. Fructose
☐ c. Pentose
☐ d. NADPH
 3. In Glycolysis, the synthesis of ATP is produced by.....
☐ a. Phosphohexose isomerase
☐ b. Glucokinase
☒ c. Pyruvate kinase
☐ d. Aldolase
 4. Which of the following is a high energy molecule in glycolysis?
☐ a. dihydroxyacetone-P
☐ b. glyceraldehydes 3- P
☒ c. 1,3-biphosphoglycerate
☐ d. 2 phosphoglycerate
 5. Which of the following molecules is synthesized in fermentation?
☐ a. FAD^+
☐ b. FADH
☒ c. NAD^+
☐ d. NADH
 6. Which of the following enzymes is important in glycogenolysis?
☒ a. Phosphorylase
☐ b. Hexokinas
☐ c. Glucokinase
☐ d. Phosphatase
 7. Which glucose transporter is active in muscle tissues?
☐ a. GLUT-1
☐ b. GLUT-2
☐ c. GLUT-3
☒ d. GLUT-4
 8. All the following enzymes are important in gluconeogenesis except:
☐ a. PEPCK
☐ b. Fructose 1,6 bisphosphates'
☐ c. Glucose 6 phosphatase
☒ d. Phosphorylase

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9. How many pyruvate molecules are required for gluconeogenesis?
- a. 1
 - ☒ b. 2
 - c. 3
 - d. 4
10. Where does TCA cycle take place?
- a. Cytosol
 - ☒ b. Mitochondria
 - c. Nucleus
 - d. Cell membrane
11. Oxidation of one glucose molecule in glycolysis and TCA cycle yields how many ATP molecules (NADH=2.5, FADH=1.5)?
- a. 26-28
 - ☒ b. 30-32
 - c. 34-36
 - d. 38-40
12. The end product of glycolysis under AEROBIC conditions is.....?
- a. Fructose-6-phosphate
 - b. Pyruvic acid
 - c. Oxaloacetic acid
 - ☒ d. Lactic acid
13. Which tissue plays a central role in processing and distributing dietary nutrients during the absorptive state?
- a. Liver
 - b. Adipose tissue
 - ☒ c. Muscle
 - d. Brain
14. What is the primary source of cytosolic acetyl coenzyme A (acetyl CoA) for fatty acid synthesis in the liver?
- a. Glucose 1-phosphate
 - ☒ b. Pyruvate
 - c. Glycerol 3-phosphate
 - d. Propionyl CoA
15. Which of the following is a result of decreased insulin levels during fasting?
- ☒ a. Increased glucose uptake by tissues
 - b. Increased fatty acid oxidation
 - c. Increased glycogen synthesis
 - d. Increased amino acid uptake by tissues
16. Glucagon stimulates which process in the liver?
- a. Glycogen synthesis
 - b. Glucose uptake
 - c. Lipogenesis
 - ☒ d. Glycogenolysis



17. All the following molecules are present in the path of cholesterol synthesis, except

- a. Geranyl
- b. Farnesyl
- c. Lanosterol
- d. Dolichol

18. What is the role of citrate in fatty acid synthesis in the cell?

- a. transfer of acetyl group from mitochondria to cytosol
- b. transfer of CO₂ from mitochondria to cytosol
- c. transfer of reducing equivalents from mitochondria to cytosol
- d. transfer of acetyl group from cytosol to mitochondria

19. How many ATP or its equivalent is produced from the aerobic and complete catabolism of oleic acid (Δ^9 18:1) (NADH=3, FADH=2)?

- a. 144
- b. 146
- c. 148
- d. 151

20. All the following factors are involved in chylomicron metabolism, except:

- a. LPL
- b. Apo B100 (LDL)
- c. HDL-C
- d. Apo CII

21. For the synthesis of palmitic acid, all of the following are needed, except:

- a. Carnitine
- b. Acetyl-CoA
- c. ATP
- d. NADPH

22. From beta oxidation of lauric acid in mitochondria, all the following compounds are produced, except:

- a. Malonyl CoA
- b. Acetyl CoA
- c. FADH₂
- d. NADH

23. In familial hypercholesterolemia, there is a mutation in which one of the following genes?

- a. LDL Receptor
- b. Apo E
- c. LPL
- d. Apo B100

24. Which of the following complexes in the electron transport chain is not involved in pumping protons across the mitochondrial membrane?

- a. Complex I
- b. Complex II
- c. Complex III
- d. Complex IV

25. Which compound inhibits ATP synthase by directly binding to it and preventing proton translocation?
- a. Rotenone
 - b. Antimycin A
 - c. Cyanide
 - d. Oligomycin
26. Which of the following compounds uncouples oxidative phosphorylation by dissipating the proton gradient across the mitochondrial membrane?
- a. Rotenone
 - b. Antimycin A
 - c. 2,4-Dinitrophenol (DNP)
 - d. Cyanide
27. What is the main purpose of the proton gradient established by the electron transport chain?
- a. To provide energy for ATP synthase to synthesize ATP
 - b. To directly convert ADP to ATP
 - c. To transport oxygen into the mitochondria
 - d. To reduce NAD⁺ to NADH
28. What is the final product of β -oxidation of odd-chain fatty acids?
- a. Acetyl-CoA
 - b. Propionyl-CoA
 - c. Butyryl-CoA
 - d. Malonyl-CoA
29. All statements regarding 3-OH-3 methyl glutaryl CoA are true except
- a. It is formed in the cytoplasm
 - b. Required in ketogenesis
 - c. Involved in synthesis of Fatty acid
 - d. An intermediate in cholesterol biosynthesis
30. Which of the following is the rate-limiting step of fatty acid oxidation and is also inhibited by malonyl CoA?
- a. Thiolase
 - b. Carnitine Palmitoyl transferase I
 - c. Acyl CoA Dehydrogenase
 - d. Acyl CoA Carboxylase
31. Which type of hyperlipidemia is associated with defects in ApoE protein variants?
- a. Type 1
 - b. Type 2b
 - c. Type 3
 - d. Type 4
32. Adipose tissue lacks
- a. Hormone-sensitive lipase
 - b. Glycerol kinase
 - c. cAMP-dependent protein kinase
 - d. β -oxidation enzymes



33. Which of the following is not true regarding Ribonucleotide reductase?

- a) It uses Thioredoxin as a coenzyme
- b) Converts NDPs to dNDPs
- ☒ c) It is inhibited in ADA deficiency
- d) Plays a major role in dTMP synthesis

34. donates its phosphate to ADP to yield ATP during muscle exertion

- a. Creatine phosphate
- ☒ b. Creatine kinase
- c. Guanidoacetate
- d. Arginine glycine amidinotransferase

35. Tophus is a clinical sign of which condition?

- a. hypouricemia
- ☒ b. hyperuricemia
- c. hyperbilirubinemia
- d. pre-hepatic jaundice

36. Jaundice is caused by all of the following, except for...?

- ☒ a. High amount of fatty acid
- b. Impaired conjugation of bilirubin
- c. obstruction of biliary tree
- d. Excess RBC lysis

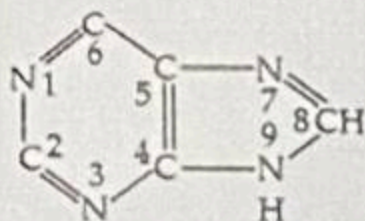
37. Which of the following is incorrectly paired regarding the disease and its indicator?

- a. Creatine => muscle disease
- ☒ b. Creatinine => kidney disease
- c. Bilirubin => Jaundice
- d. Urea => Uricemia

38. In Lesch-Nyhan syndrome

- ☒ a. There is a defect or lack in the HGPRT enzyme
- b. There is an excess in the HGPRT enzyme
- c. HGPRT enzyme is not involved in the disease
- d. None of the above

39. According to the biosynthetic origin of purine ring, nitrogen atom N1 originates from?



- a. Glutamine
 - b. Glycine
 - ☒ c. Aspartic acid
 - d. Formate
40. What is a primary source of Non-Protein Nitrogen (NPN) in ruminant diets?
- ☒ a. A. Cellulose
 - ☒ b. B. Urea
 - c. C. Starch
 - ☒ d. D. Glucose
41. Histamine, a chemical mediator of allergies and anaphylaxis, is synthesized from amino acid Histidine by which of the following process?
- a. Decarboxylation
 - ☒ b. Demethylation
 - c. Hydrogenation
 - d. Oxygenation
42. Which of the following statements about the urea cycle is correct?
- a. Argininosuccinate is lysed to urea and ornithine in the urea cycle.
 - b. Carbamoyl phosphate supplies both of the nitrogen atoms of urea in the urea cycle.
 - ☒ c. Arginine is hydrolysed to urea and ornithine in the urea cycle.
 - d. The formation of urea from the urea cycle yields energy.
43. Which of the following statements about transamination reactions is correct?
- a) Transamination reactions involve ATP hydrolysis.
 - b) Transamination reactions are irreversible.
 - c) Transamination reactions require NAD^+ or NADP^+ .
 - ☒ d) Transamination reactions require pyridoxal-5'-phosphate (PLP).



44. Which of the following amino acids cannot provide a substrate for gluconeogenesis?

- a) Leucine
- b) Tryptophan
- c) Histidine
- d) Isoleucine

45. Which of the following transport system is responsible for transferring dipeptides from the intestinal lumen to enterocytes?

- a) K⁺-linked secondary transport
- b) Facilitated diffusion
- c) Na⁺-linked secondary transport
- d) H⁺-linked transport system

46. Which of the following enzyme is deficient in alkaptonuria?

- a) Homogentisate dioxygenase
- b) p-Hydroxyphenyl- pyruvate dioxygenase *pHp d*
- c) Phenylalanine hydroxylase
- d) Tyrosinase

47. Which of the following amino acids are involved in the transfer of ammonia from tissues to the liver?

- a) glutamate, glutamine
- b) aspartate, alanine
- c) aspartate, glutamate
- d) Alanine, glutamine

48. Identify the amino acid that participates in methyl group transfer:

- a) Lysine
- b) Serine
- c) Valine
- d) Methionine

49. Which of the following is the only plasma enzyme normally found in urine?

- a) Acid Phosphatase
- b) Amylase
- c) Nucleotidase
- d) Aminotransferase

50. Which of the following are the most important cause of increased alkaline phosphatase activity in serum?

- a) liver disease, heart attack
- b) heart attack, prostate disease
- c) liver disease, bone disease
- d) bone disease, prostate disease

51. Serum nonfunctional enzymes are all, except:

- a) Alkaline phosphatase
- b) Acid phosphatase
- c) Lipoprotein lipase
- d) Gamma glutamyltranspeptidase

35-12:
-12
-1
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52. Which enzyme in the body is measured to diagnose rhabdomyolysis?
- a) Lipase
 - b) Nucleotidase
 - ☒ c) Creatine kinase
 - d) Glutamyltransferase